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Finance and the supply of housing quality[☆]

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ABSTRACT

I show how financial intermediaries affect rental housing quality and affordability by supplying real estate investors with financing for quality improvement projects (i.e., renovations). First, I document a historic surge in improvement activity since the Great Recession. Then, using exogenous variation generated by a 2015 change in regulatory capital requirements, I find that a reallocation of bank credit toward improvement projects accounts for 24% of quality improvements since 2015. The shock increases the supply of high-quality apartments and lowers their rent. However, it raises the average apartment's rent and accounts for 32% of historically high rent growth over 2015–16.

1. Introduction

Over one-third of U.S. households rent their home, and many of these households have experienced historically high housing costs since the Great Recession.[1] These observations have ignited policy discussion about housing affordability, and they have inspired a new research agenda focused on the rental housing market (e.g., Diamond et al., 2019; Howard and Liebersohn, 2020; Favilukis et al., 2019; Molloy et al., 2020; Gete and Reher, 2018). Missing from this discussion is the fact that improvements to rental housing quality (i.e., renovations) have also surged since the Recession, as I document shortly. This new observation is central to discussions about housing affordability, since improvements reduce the supply of relatively inex-

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[1] According to the Housing Vacancy Survey, 37% of households were renters in 2016, rising to 50% in urban metro areas such as Los Angeles and New York City. The median rent-to-income ratio reached 30% in 2015, its highest level since the 1980s (Gete and Reher, 2018).

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pensive housing units by transforming them into relatively expensive ones.

I document a recent surge in improvements to rental housing quality, a new fact with direct implications for housing affordability. Then, I show how financial intermediaries have contributed to this surge by reallocating financing toward quality improvements and away from other types of residential investment. Using an unintended regulatory spillover shock from the application of Dodd-Frank bank capital requirements to apartment loans, I find that credit supply accounts for 24% of improvement activity since 2015. Tracing the effects downstream, this shock lowers rent growth on high-quality apartments by increasing their supply, while increasing average rent growth as more apartments become high quality. These findings exemplify how financial intermediaries can function as suppliers of housing quality and, through this role, affect housing affordability.

In more detail, I first use a variety of proprietary and public datasets to show how improvement activity in the rental housing sector has rebounded to historically high levels since the Great Recession. This surge has been accompanied by an increased share of rental housing units in the upper segments of the quality distribution along with depressed real rent growth in these segments. By contrast, average rent growth remains high by historical standards. Together, these facts are consistent with an outward shift in the supply of high-quality rental housing due to increased improvement activity. This conjecture motivates me to study an exogenous shock to the supply of financing for improvement projects.

I study a credit supply shock for apartment improvements generated by High Volatility Commercial Real Estate (HVCRE) bank capital requirements. These requirements were introduced in 2015 as part of the Dodd-Frank Act's implementation of Basel III international standards, and, thus, they were not motivated by specific features of the U.S. real estate market. HVCRE regulation assigned a 50% lower regulatory risk weight to loans secured by improvements on income-producing properties relative to loans for construction. Thus, this regulatory shock introduced a wedge in the cost of capital for different loan types, incentivizing banks to transfer credit to improvement projects from construction. I verify that banks respond in this way using a triple difference-in-difference methodology that compares the allocation to improvement projects by banks (i.e., "treated lenders") and specialty nonbank lenders within the apartment loan market. The results show how banks and nonbanks follow parallel trends leading up to the introduction of HVCRE regulation, after which banks reallocate loanable funds to improvements. This lender-level finding supports the shock's validity by serving as a "first-stage."

My baseline exercise is a county-level difference-in-difference research design, where a county's treatment exposure is defined by its historical reliance on bank as opposed to nonbank lending. I carry out this research design using a novel dataset on both portfolio and securitized loans originated by both bank and nonbank lenders. As in the lender-level analysis, bank-reliant counties (i.e., "treated counties") and nonbank-reliant counties exhibit parallel trends until the introduction of HVCRE regulation. After the shock, however, bank-reliant counties exhibit significantly higher real improvement activity, based on a variety of outcome measures. The effects are stronger in counties where real estate investors have fewer sources of credit and where renters appear more willing to pay for housing quality (e.g., higher-income renters), suggesting that the shock relaxes constraints on investors' demand for improvement financing.

The baseline results are internally valid insofar as bank and nonbank-reliant counties do not differ in unobserved ways that would affect improvement activity after the 2015 shock. I conduct numerous robustness tests, all of which support the baseline results' internal validity. These include: replicating the baseline analysis with an alternative dataset; controlling for changes in demand from the Government Sponsored Enterprises (GSEs); dropping improvements by unconstrained borrowers; checking that the results do not confound other specific Dodd-Frank regulations; controlling for a battery of county characteristics and state-by-year fixed effects; investigating differences in loan specialization between banks and nonbanks; and evaluating bias from undersampling small regional banks that specialize in construction. I also estimate a property-level difference-in-difference equation, which allows me to rely on a very weak identification assumption that is immune to unobserved characteristics of a county. The property-level results imply that HVCRE regulation raises a property's annual probability of an improvement by 46% (1.2 pps).

Turning downstream, I trace the shock's effects through to commonly used measures of housing affordability. Consistent with the lender-level reallocation, I find that HVCRE regulation reduces county-level apartment construction. Yet, despite this negative effect on the overall supply of apartments, the shock actually increases the supply of high-quality apartments, since it generates improvements that transform low-quality apartments into high-quality ones. This increase in supply is accompanied by lower rent growth on high-quality apartments. By contrast, the shock significantly raises rent growth on the average apartment, reflecting the joint contribution of higher average quality and reduced overall supply. In particular, the shock raises the share of households with a rent-to-income ratio above policymakers' standard definition of "cost-burdened" (e.g., JCHS (2017)). The overall welfare content of this outcome is unclear because higher rent also reflects better quality. In distributional terms, however, the shock likely favors higher-income households insofar as housing quality is a normal good.

I assess the aggregate impact of HVCRE regulation using methods from the applied macroeconomics literature (e.g., Chodorow-Reich, 2014). Correspondingly, I find that the shock accounts for 24% of quality improvements over 2015–16 and 32% of rent growth, in partial equilibrium. To assess the plausibility of this large effect, I perform a hedonic adjustment with an auxiliary dataset used by statistical agencies. This adjustment calculates the share of rent growth attributable to quality improvements, without taking a stance on the shocks generating these improvements. Accordingly, I find that improvements due to any shock account for 70% of observed real rent growth following the Great Recession. Therefore, the large estimated effect of the specific HVCRE shock that I study is indeed plausible.

The remainder of the paper is organized as follows. I conclude this section by situating my contribution within the related literature. Section 2 describes the paper's data. Section 3 documents several new stylized facts and clarifies my research hypothesis. Section 4 describes the regulatory shock. Section 5 contains my core, county-level analysis. Section 6 assesses internal validity. Section 7 studies implications for housing affordability. Section 8 studies aggregate implications. Section 9 concludes. The online appendix contains additional material.

This paper makes three contributions to the literature. First, the results show how constraining the supply of market-rate (i.e., non-subsidized) housing reduces the supply of more affordable housing through a novel channel, namely, a shift from producing high-quality housing through new construction to producing it through the improvement of low-quality housing. This quality improvement channel works in parallel to the standard supply-and-demand channel (e.g., Asquith et al., 2019), and it is conceptually similar to the process of downward filtering (e.g., Rosenthal, 2014). In fact, quality improvements themselves constitute upward filtering, and so the results help explain cross-sectional variation in such upward filtering recently documented by Liu et al. (2020).

Second, I show how financial intermediaries play a critical role in urban change by providing financing to real estate investors. In particular, the results suggest incorporating financial frictions into equilibrium models of gentrification (e.g., Guerrieri et al., 2013; Couture et al., 2020). Moreover, my focus on the supply of housing quality complements research on households' demand for living in different quality segments (e.g., Landvoigt et al., 2015; Piazzesi et al., 2020) or improving their own home (Benmelech et al., 2021). In policy terms, a number of recent papers have studied how urban policies such as tax credits or rent stabilization affect rental housing markets (e.g., Favilukis et al., 2019; Diamond and McQuade, 2019; Diamond et al., 2019), and this paper shows how the rental market is also affected by upstream financial regulation. Lastly, a large literature has studied the effect of financial markets on housing markets in the owner-occupied sector, and this paper is among a smaller set to study that effect within the rental market (e.g., Greenwald and Guren, 2020; Gete and Reher, 2018).

Third, viewing housing quality as a good and improvement projects as a technology used to produce it, my findings support a large and diverse literature showing how financial intermediaries affect the level and allocation of productive resources. By estimating these effects using a regulatory spillover shock, I contribute to a literature on the unintended impact of Dodd-Frank regulations on housing markets (e.g., Defusco et al., 2020; D'Acunto and Rossi, 2021).[2] Of course, the unintended effect of HVCRE regulation on housing affordability must be weighed against the intended effects on financial stability and the banking sector, as studied by Glancy and Kurtzman (2018).

2. Data

My core analysis relies on two datasets, both of which are provided by Trepp LLC. Together, they cover a representative panel of apartment properties over 2010–16. I also rely on multiple auxiliary datasets to assess the robustness of my core findings. All of these datasets are described in detail in Appendix A, and I provide a concise overview below.

Before doing so, I introduce some important terminology that will clarify the rest of the paper. I define "quality" as a structural feature of a shelter. I will use "improvement" as the general term for an increase in quality, which will include large-scale projects that require the inhabitant to vacate (e.g., renovations) as well as small-scale ones (e.g., installing an air conditioner). Most of my analysis takes place in the apartment sector, and I will use the word "apartment" to refer to the individual housing unit and "property" to refer to the entire set of units under common ownership. Finally, the "borrowers" in my analysis are professional real estate investors, not owner-occupants seeking home improvement loans.

2.1. Core datasets

The first source of data is Trepp's Anonymized Loan Level Repository (T-ALLR) dataset. The T-ALLR dataset contains regularly updated information on bank-originated loans secured by apartment properties. Importantly for the purposes of this paper, the T-ALLR dataset consists of loans that remain on the bank's balance sheet (i.e., portfolio loans). Data on portfolio loans are considered highly confidential in the U.S. and are thus difficult to acquire, but incorporating them into the analysis is important given my focus on the effects of bank capital requirements in Section 5.

While attractive due to its coverage of portfolio loans, the T-ALLR dataset does have three drawbacks. First, the dataset only includes loans originated by banks, whereas my identification strategy will require information about both bank and nonbank lenders. Second, the raw data cover only 10% of U.S. counties, or 52% on a population-weighted basis. Third, I observe whether a loan finances construction, but, among the remaining loans that are secured by income-producing properties, I cannot explicitly identify those that finance an improvement.

Given these constraints, I also incorporate Trepp's T-Loan dataset into my core analysis. The T-Loan dataset contains regularly updated information on apartment loans that have been securitized as commercial mortgage backed securities (CMBS). Consequently, the T-Loan dataset includes loans originated by both bank and nonbank originators, whom I will simply call "lenders" to maintain a

[2] Relatedly, I contribute to a literature on the rise of shadow banks by providing new evidence that capital requirements affect nonbanks' market share (e.g., Buchak et al., 2018; Kim et al., 2018; Fuster et al., 2019; Irani et al., 2021; Chernenko et al., 2020; Ganduri, 2020; Gete and Reher, 2021).

consistent terminology. The largest of these lenders are listed in Appendix Table A1. In terms of breadth, the T-Loan dataset covers 90% of population-weighted counties, and it includes a rich set of property-level variables, such as rent, size, occupancy, and, importantly, the history of renovations on the property.

Comparing the two sources of data, bank-originated loans observed in the T-ALLR dataset encode stronger regulatory incentives than those in the T-Loan dataset, since securitized loans only incur capital requirements during the warehouse period or through a risk-retention ratio, as discussed in Section 4.1. In this sense, the T-ALLR dataset is superior. On the other hand, the T-Loan dataset has wider geographic coverage and richer property-level information. Given these tradeoffs, I combine the two Trepp datasets in my core analysis.

2.2. Auxiliary datasets

I draw on multiple auxiliary datasets to ensure that my core findings from the combined Trepp dataset are robust. The two most significant of these auxiliary datasets are a dataset on apartment transactions from Real Capital Analytics (RCA) and the Census' American Housing Survey (AHS) dataset.

The RCA dataset covers transactions on apartment properties over 2009–16. I observe the lender and loan size associated with the transaction, the history of renovations on the transacting property, and its location. Together, these variables enable me to replicate my core analysis with the RCA dataset, which, as I will show in Table 3, leads to similar results. While useful for evaluating robustness, the RCA dataset contains a limited set of outcome variables, and it does not cover improvements on non-transacting properties because the raw data are collected from transactions (CREDA, 2017). Therefore, given this paper's focus, I do not use the RCA dataset as my primary source of data.

The AHS dataset is nationally representative and contains information about a housing unit's rent, the demographic profile of the occupant, and granular information about the housing unit's physical features. The granularity and representativeness of the AHS dataset make it ideal for the quality adjustment exercise in Section 8.1, where I assess the plausibility of the core results' aggregate implications. However, the AHS dataset has rather coarse geographic information, as I only observe a housing unit's metropolitan statistical area (MSA) and only for a subset of 43% of MSAs. In addition, the AHS dataset's panel structure broke because of a redesign in 2015, and so I cannot follow the same housing unit throughout the entirety of my core sample period of 2011–16. For these reasons, the AHS dataset complements my core dataset but cannot substitute for it.

3. Stylized facts

I motivate the paper by documenting two stylized facts about rental housing quality and affordability, shown in Fig. 1. Panel (a) documents a surge in quality improvement activity since the Great Recession by plotting the percent of apartments that are renovated each year, based on the T-Loan dataset. This annual probability of renovation vigorously recovered from its 2008 low and surpassed its pre-Recession high by 2014. Appendix Figure A1 replicates this finding using three separate measures of improvement activity: aggregate investment in residential improvements, based on the U.S. Fixed Assets Accounts; the share of bank portfolio loans that proxy for financing an improvement, based on the T-ALLR dataset; and the probability an apartment transaction is followed by a renovation, based on the RCA dataset.

Panel (b) of Fig. 1 documents a negative cross-sectional correlation between housing quality and rent growth. Using the AHS dataset, I sort renters into quintiles based on log real income relative to the MSA-year average, a proxy for quality segment. Next, I plot annualized real apartment rent growth for each segment over 2011–17. While real rent grew at least 2.9% per year for the bottom three quintiles, it only grew at a rate of 0.9% for the top quintile. This pattern is robust to a zip code-level analogue based on Zillow's apartment rent index, also shown in Appendix Figure A1.

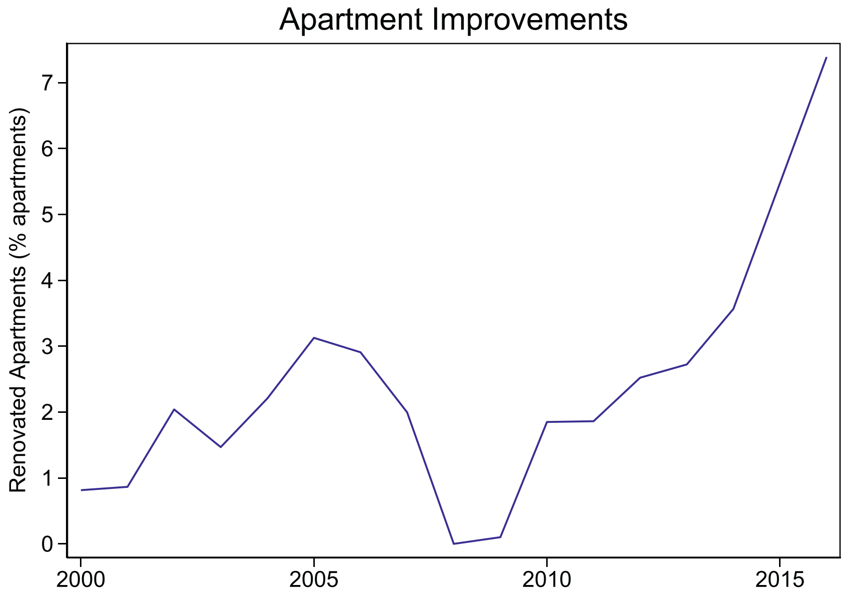
To the best of my knowledge, the trends shown in Fig. 1 have not yet been documented in the literature. My goal in the remainder of the paper is to assess how shifts in the supply of financing have contributed to them. Before doing so, I briefly describe my principal research hypothesis.

3.1. Research hypothesis

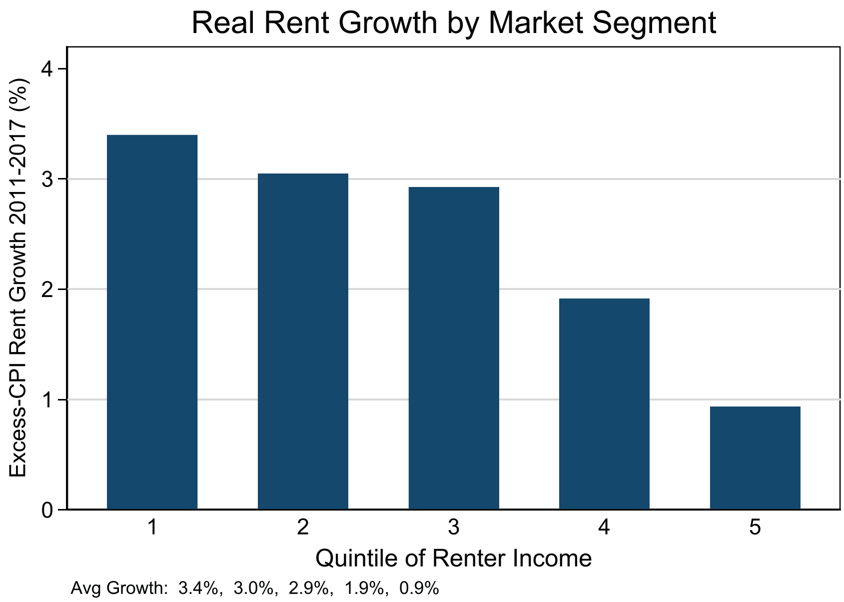
To fix ideas, suppose there is a distribution of rental housing quality, where, as mentioned in Section 3, quality is defined as a structural feature of a shelter (e.g., air conditioning). An improvement project raises the quality of a housing unit, thereby moving it from a lower segment of the quality distribution to a higher segment. Real estate investors own the rental housing stock and perform improvement projects. To do so, they rely on outside financing, which is consistent with the observation that 70% of improvements occur on a mortgaged property according to the 2015 Rental Housing Finance Survey (RHFS).

Financial intermediaries supply real estate investors with financing to perform improvement projects. Consider a shock that incentivizes intermediaries to supply more improvement financing, such as the regulatory spillover shock described in Section 4.1. The increase in supply can take two forms: a relaxation of borrowing constraints for constrained investors (i.e., borrowers), or lower interest rates for unconstrained investors, holding credit risk fixed. Table 6 and Appendix Table A5 will respectively document these two channels. Both channels imply an increase in quality improvement activity, as Table 2 will show. The specific supply shock I study entails a reallocation of financing toward improvements and away from construction due to a wedge in regulatory capital requirements, and so I verify that it also reduces construction activity in Table 7.

The increase in quality improvement activity affects housing affordability by expanding the supply of high-quality housing at the expense of low-quality housing, as I will also show in Table 7. In particular, high-quality housing units command a premium in rent over low-quality



(a) Apartment Improvements



(b) Real Rent Growth by Market Segment

Fig. 1. Facts about rental housing quality and affordability. *Note:* This figure documents two new stylized facts about rental housing quality and affordability. Panel (a) plots the percent of apartments renovated each year, based on Trepp's T-Loan dataset. Panel (b) plots real (i.e., excess-CPI) growth in average apartment rent by the renter's income quintile, based on the Census' AHS dataset. Renters are sorted into quintiles by their log real income relative to the MSA-year mean and weighted using AHS survey weights.

units, reflecting households' willingness to pay for quality. Therefore, by raising the average housing unit's quality, the increase in quality improvement activity also raises the average housing unit's rent, as Table 8 will show. However, households must accommodate the increase in the supply of high-quality housing units, which necessitates a reduction in the quality premium. Thus, while the average housing unit's rent increases, the rent on high-quality units falls, as I will also show in Table 8.

4. Identification

My goal is to estimate the effect of financial supply on quality improvement activity, and, subsequently, to trace this effect through to measures of housing affordability. I identify the effect through a regulatory spillover shock that increases the supply of bank credit for apartment improvements. In this section, I describe the shock and provide graphical evidence of its effect.

4.1. High volatility commercial real estate

In January 2015, U.S. bank regulators increased the regulatory capital risk weight on certain apartment loans, called High Volatility Commercial Real Estate (HVCRE) loans, from 100% to 150%. This means that banks must reserve at least $\$1.50 \times K$ in equity capital for every \$1 of HVCRE credit, where K is the regulatory minimum capital ratio (e.g., 6%). HVCRE loans are for the "development or construction of real property," which I will simply refer to as "construction" (U.S. Code §1831bb(b)(1)(A)).[3] By contrast, loans for "improvements to existing income-producing real property" (U.S. Code §1831bb(b)(2)(C)) were not subject to this increase and retained the substantially more modest weight of 100%. As its name implies, HVCRE regulation affects loans secured by apartment properties because apartments are considered "commercial real estate," but it exempts loans secured by single-family homes (U.S. Code §1831bb(b)(2)(A)).[4] Importantly, the particular HVCRE risk weights were chosen to conform with international standards set by the Basel III Accords, and, thus, they are unrelated to specific features of the U.S. real estate market.

If the Modigliani-Miller theorem fails, then HVCRE regulation can increase the supply of bank credit for improvement projects based on the following logic. First, if it is sufficiently costly for banks to raise equity capital compared to debt, then regulatory capital requirements bind. A large literature summarized by Dagher et al. (2016) has found this to be the empirically relevant case during periods of regulatory transition. Consequently, banks must raise $\$1.00 \times K$ in equity capital for a \$1 loan that finances an improvement project, relative to $\$1.50 \times K$ for one that finances construction. To minimize its regulatory burden while maintaining the same exposure to commercial real estate projects as a whole, a bank can respond by reallocating loanable funds toward improvements and away from construction. Table 6 will document this behavior formally, and, consistent with theory, Appendix Table A5 will show how it is especially pronounced among poorly capitalized banks. While, in principle, a bank could respond by simply originating fewer construction loans, Table 6 will show how banks actually increase their improvement lending, thereby maintaining their total exposure to commercial real estate projects. These predictions are similar to those conjectured by Greenwood et al. (2017).

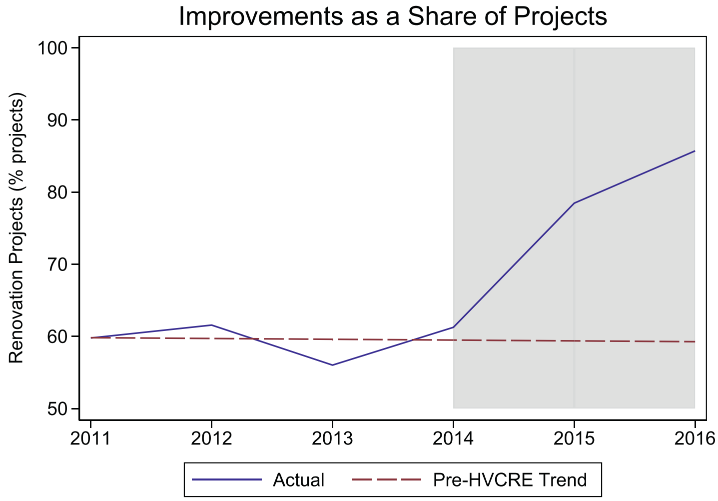
The previous logic applies most strongly to portfolio loans, since they remain on a bank's balance sheet and so necessitate that the bank reserve equity capital for an extended period. In the case of securitized loans, a bank must reserve equity capital throughout the loan's warehouse period, during which it is booked as held-for-sale and subject to standard risk weights (FFIEC, 2015). The average warehouse period in the T-Loan dataset is six months, which is long enough to incentivize some form of regulatory arbitrage. After securitization, capital requirements still bind through the retained portion of a loan. The retained portion is typically 5% due to risk retention rules (e.g., Flynn et al., 2020; Willen, 2014), although loans purchased by the Government Sponsored Enterprises (GSEs) are exempt from these rules. For these reasons, securitized loans still encode regulatory incentives, although more weakly than portfolio loans. Therefore, I incorporate both loan types into my core analysis.

Mapping to a difference-in-difference setup, banks constitute "treated lenders," whereas nonbanks are not subject to capital requirements and so constitute "control lenders" (e.g., Buchak et al., 2018). I define the post-2015 period as the "treatment period," corresponding to the period after which HVCRE regulation was introduced.[5] While the regulation's announcement occurred in 2013 as part of the U.S. implementation of Basel III, lenders were sufficiently confused about the regulation's details that they did not adjust ex-ante (e.g., MBA (2018)). As evidence of this confusion, federal regulators issued a clarificatory statement in March 2015 (FDIC, 2015). Indeed, the graphical evidence

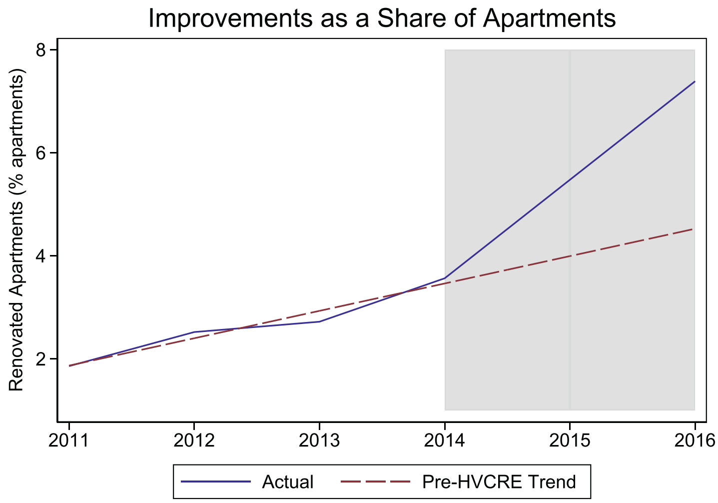
[3] In addition, the loan must satisfy any of the following underwriting conditions met by most construction projects (Chandan and Zausner, 2015): the loan-to-value (LTV) ratio exceeds 80%; the terms allow capital withdrawals; or the borrower's contributed capital is less than 15% of the project's as-completed value.

[4] The institutional details described in this section are based on the codified definition of HVCRE loans, which did not appear until after HVCRE regulation became effective in 2015. In addition, the U.S. Senate modified HVCRE regulation after this paper's period of analysis as part of its Economic Growth, Regulatory Relief, and Consumer Protection Act (S. 2155).

[5] In studying the effect of HVCRE regulation on interest rates, Glancy and Kurtzman (2018) define the "treatment period" as the period following the regulation's announcement (i.e., post-2013). However, in addition to the reasons described in the text, I do not use this definition because my setting differs from Glancy and Kurtzman (2018). First, I focus on the quantity of loans originated rather than the interest rate conditional on origination, and interest rates are presumably forward-looking. Second, whereas Glancy and Kurtzman (2018) only study portfolio loans, my data also include securitized loans, for which there is less incentive to adjust ex-ante because capital requirements primarily bind during the warehouse period. Lastly, loans originated before 2015 were grandfathered once HVCRE regulation was codified, although grandfathering occurred only after codification.



(a) Improvements as a Share of Projects



(b) Improvements as a Share of Apartments

Fig. 2. Aggregate improvement activity around the HVCRE regulatory shock. *Note:* This figure plots aggregate quality improvement activity around the HVCRE regulatory shock. Panel (a) plots the number of renovation projects divided by the sum of renovation projects and construction projects, all within the apartment sector. Panel (b) plots the percent of apartments renovated each year. The gray region denotes the period when HVCRE regulation is in place. Data are from Trepp's T-Loan dataset.

presented in the next subsection shows how lenders do not adjust ex-ante, and so I take 2015 as the shock year.

4.2. Graphical evidence

I begin with two pieces of graphical evidence supporting the interpretation of HVCRE regulation as a positive shock to the supply of financing for improvement projects. First, panel (a) of Fig. 2 plots the time series of the aggregate distribution between improvement and construction projects. The distribution is stable leading up to the HVCRE shock, after which it tilts sharply toward improvements. Panel (b) shows how this shift in aggregate project composition comes with a sharp increase in quality improvement activity relative to its pre-HVCRE trend, as measured by the share of apartments that are renovated each year. The figure is based on the T-Loan dataset because of its breadth and depth, but Appendix Figure A1 documents similar dynamics using the T-ALLR and RCA datasets.

Second, Fig. 3 verifies that banks (i.e., "treated lenders") increase their allocation to improvements after the HVCRE shock. The figure plots the year-by-year average difference in log loan originations between renovation vs. construction projects between bank vs. nonbank lenders, based

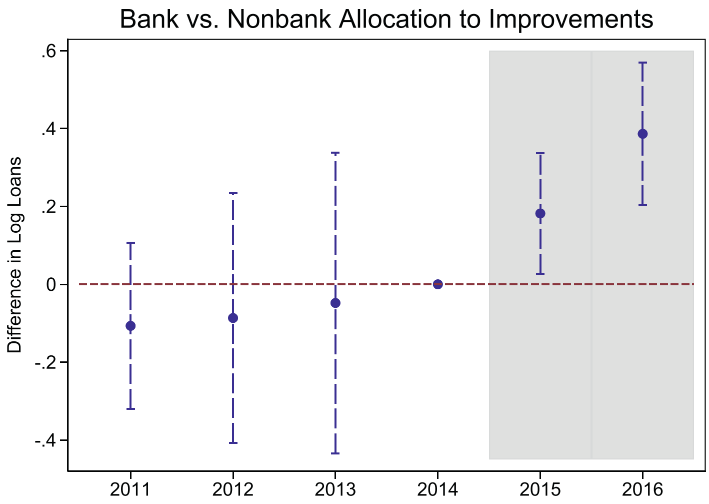


Fig. 3. Improvement lending around the HVCRE regulatory shock. *Note:* This figure plots the year-by-year average difference in log loan originations between renovation and construction projects between bank and nonbank lenders, which allows for an inspection of pre-trends in relative improvement lending. The reference year is 2014. The difference is demeaned by a lender-year fixed effect, a loan type-year fixed effect, and a loan type-bank fixed effect, as in Table 6. The gray region highlights the period when HVCRE regulations are in place. Brackets are a 95% confidence interval with standard errors clustered by lender and year. Data are from Trepp's T-Loan dataset.

on the T-Loan dataset in which I observed the lender's identity. There is no ex-ante adjustment before the HVCRE shock (i.e., no pre-trend), substantiating the claim made in the previous subsection. After the shock, however, banks significantly tilt their portfolio toward improvement loans relative to nonbanks. This shift in lending behavior constitutes the HVCRE shock's "first stage," and I perform numerous additional tests in Section 6.8 to verify the validity of this first-stage effect. I refer to the lender-level effect as a "first stage," since it partly captures shifts in bank vs. nonbank market share and does not necessarily imply an increase in real quality improvement activity. Therefore, I conduct my core analysis at the county level, as now described.

5. Effect on quality improvements

My core analysis is a county-level difference-in-difference research design in which I estimate the effect of HVCRE regulation on quality improvement activity. By definition, quality improvements raise the supply of more-expensive housing while reducing the supply of less-expensive housing. Thus, the results in this section bear directly on discussions about housing affordability, and I will use the same research design introduced below to make this link explicit in Section 7.

5.1. Baseline specification

Aggregating to the county-level enables me to assess the real effects of HVCRE regulation, and so I estimate the following regression equation over 2011–16,

$$Y_{c,t} = \beta(Bank\ Share_c \times Post_t) + \alpha_c + \alpha_t + \gamma X_{c,t} + u_{c,t}, \quad (1)$$

where c and t index counties and years; $Bank\ Share_c$ measures banks' share of apartment loan balances in 2010, as opposed to nonbanks' share; $Post_t$ indicates if t is greater than or equal to 2015; $Y_{c,t}$ is one of several measures of quality improvement activity; and the controls in $X_{c,t}$ include contemporaneous measures of local housing and credit demand and, in some specifications, state-year fixed effects.[6]

Interpreting Eq. (1), the "treatment" is the introduction of HVCRE regulation in 2015, and "treated counties" are those that have historically relied on bank financing, as parameterized by $Bank\ Share_c$. I measure $Bank\ Share_c$ in three ways. First, I calculate

$$Portfolio\ Bank\ Share_c = \frac{Bank\ Portfolio\ Balances_c}{Bank\ Portfolio\ Balances_c + All\ Securitized\ Balances_c}, \quad (2)$$

where $Bank\ Portfolio\ Balances_c$ is the value of bank portfolio loan balances in 2010, based on the T-ALLR dataset; and $All\ Securitized\ Balances_c$ is the value of bank and nonbank securitized loan balances in 2010, based on the T-Loan dataset.[7] Since 84% of apartment loans are either securitized or held on banks' portfolios (Rosengren, 2017),

[6] County controls are: log real income per capita for the surrounding MSA, based on data from the Bureau of Economic Analysis; log winter storms, based on data from the National Oceanic and Atmospheric Association; and the principal-weighted averages of the loan-to-value ratio, debt service coverage ratio, adjustable rate mortgage share, and 60+ day delinquency rate on existing apartment loans, based on the T-Loan dataset. I follow standard practice and add one to the variable before taking the log whenever the variable can equal zero. Observations are weighted by the county's average number of apartments over 2011–16, based on the T-Loan dataset.

[7] Taking the unweighted sum of balances observed in the T-ALLR and T-Loan datasets as in the denominator of Eq. (2) leads to measurement error because neither dataset fully represents each U.S. county. I address this measurement error by reweighting securitized loan balances such that the share of loan balances held on banks' portfolios in the combined Trepp dataset matches the national accounts, as described in Appendix A.6.

Table 1
Summary statistics.

	Observations	Mean	Standard Deviation	Source of Data
Core Dataset:				
Portfolio Bank Share$_c$	3128	0.192	0.168	T-ALLR, T-Loan
Securitized Bank Share$_c$	3128	0.539	0.184	T-ALLR, T-Loan
Total Bank Share$_c$	3128	0.731	0.163	T-ALLR, T-Loan
$\log(\textit{Bank Loans}_{c,t})$	639	1.033	0.595	T-ALLR
$\log\left(\textit{Renovated Properties}_{c,t}\right)$	3128	0.154	0.377	T-Loan
$\log\left(\textit{Renovated Apartments}_{c,t}\right)$	3128	0.934	2.164	T-Loan
$\log(\textit{Renovation Value}_{c,t})$	3128	1.744	4.312	T-Loan
$\log\left(\textit{Newly-Built Properties}_{c,t}\right)$	3128	0.389	0.874	T-ALLR, T-Loan, BPS
$\Delta\log\left(\textit{Top-Quality Properties}_{c,t}\right)$	2540	0.205	0.654	T-Loan
$\Delta\log\left(\textit{Average Rent}_{c,t}\right)$	2494	0.027	0.062	T-Loan
$\Delta\textit{Cost-Burdened Share}_{c,t}$	2577	0.019	0.116	T-Loan, IRS
$\Delta\log\left(\textit{Top-Quintile Rent}_{c,t}\right)$	2057	0.014	0.049	T-Loan
GSE Share$_{c,t}$	3125	0.543	0.313	T-Loan
Transactions Dataset:				
Total Bank Share$_c$	2116	0.298	0.229	RCA
$\log\left(\textit{Renovated Properties}_{c,t}\right)$	2729	0.566	0.84	RCA
Baseline Number of Counties: 554				

Note: This table presents summary statistics of key county-level variables. Subscripts c and t denote county and year. The rightmost column lists the datasets used to construct each variable: the T-ALLR and T-Loan datasets both come from Trepp, and they respectively derive from loans held on banks' portfolios and loans securitized by bank and nonbank originators; RCA denotes the Real Capital Analytics dataset, which derives from apartment transactions; BPS denotes the Census' Building Permits Survey dataset; and IRS denotes the Internal Revenue Service's SOI Tax Stats dataset. The variables in the upper panel are defined as follows: *Portfolio Bank Share*$_c$ is the ratio of bank portfolio loan balances in 2010, based on T-ALLR, to the sum of bank portfolio loan balances in 2010, based on T-ALLR, and bank and nonbank securitized loan balances in 2010, based on T-Loan; *Securitized Bank Share*$_c$ is the ratio of bank securitized loan balances in 2010, based on T-Loan, to the same denominator as in *Portfolio Bank Share*$_c$; *Total Bank Share*$_c$ is the sum of *Portfolio Bank Share*$_c$ and *Securitized Bank Share*$_c$; *Bank Loans*$_{c,t}$ is the number of bank portfolio loans originated for purposes other than construction, based on T-ALLR; *Renovated Properties*$_{c,t}$, *Renovated Apartments*$_{c,t}$, and *Renovation Value*$_{c,t}$ are the number of renovated apartment properties, number of renovated apartment units, and total revenue of renovated apartment units, respectively, based on T-Loan; *Newly-Built Properties*$_{c,t}$ is the sum of apartment construction projects observed in the T-ALLR and T-Loan datasets or, when no construction is observed in the Trepp data, the number of apartment construction permits issued according to the BPS dataset; *Top-Quality Properties*$_{c,t}$ is the number of properties times share of apartments ranked in the top quality segment by professional property inspectors, based on T-Loan; *Average Rent*$_{c,t}$ is the average rent among all apartments, based on T-Loan; *Cost-Burdened Renters*$_{c,t}$ is the share of apartments whose rent, according to the T-Loan dataset, exceeds 30% of the zip code's average income, according to the IRS dataset; *Top-Quintile Rent*$_{c,t}$ is the average rent among apartments whose rent is above the 80th percentile within the same county-year, based on T-Loan; and *GSE Share*$_{c,t}$ is the share of securitized loan balances backed by the Government Sponsored Enterprises (GSEs) based on T-Loan. The variables in the lower panel are defined similarly but are calculated using the RCA dataset. Observations are county-years weighted by the number of apartments in the county. The sample period is 2011–16. Appendix A provides additional details.

the denominator in Eq. (2) approximates the total value of apartment loan balances in 2010. To limit sample attrition, I impute a value of zero for bank portfolio balances in counties covered by the T-Loan dataset but not by T-ALLR.

Next, I complement *Portfolio Bank Share*$_c$ with the following measure,

$$\textit{Securitized Bank Share}_c = \frac{\textit{Bank Securitized Balances}_c}{\textit{Bank Portfolio Balances}_c + \textit{All Securitized Balances}_c}, \tag{3}$$

where *Bank Securitized Balances*$_c$ is the value of bank securitized loan balances in 2010, based on the T-Loan dataset. Many specifications control for both *Portfolio Bank Share*$_c$ and *Securitized Bank Share*$_c$ to assess the explanatory power of each measure. In theory, HVCRE regulation should primarily operate through *Portfolio Bank Share*$_c$, since, as discussed in Section 4.1, banks' incentive to substitute toward improvement loans is weaker among loans securitized after a warehouse period of around 6 months.[8]

Finally, I combine Eqs. (2) and (3) to calculate the composite measure

$$\textit{Total Bank Share}_c = \textit{Portfolio Bank Share}_c + \textit{Securitized Bank Share}_c. \tag{4}$$

This composite measure parsimoniously encodes the incentives generated by HVCRE regulation and can be calculated for 79% of population-weighted U.S. counties. Table 1 provides summary statistics for all of the exposure and outcome variables used in my core analysis.

The parameter β in Eq. (1) recovers the effect of HVCRE regulation on quality improvement activity, provided the following identification assumption holds,

$$\mathbb{E}[\,\textit{Bank Share}_c \times \textit{Post}_t \times u_{c,t} | \alpha_c, \alpha_t, X_{c,t}] = 0. \tag{5}$$

In words, assumption (5) says that counties where banks have historically held a large share of the apartment loan

[8] In principle, some of the loans observed in the T-Loan dataset may have been originated by banks that intended to hold the loans on their balance sheets but later sold them to CMBS conduits. Such loans would encode the same regulatory incentives as portfolio loans observed in the T-ALLR dataset. This situation may apply to loans securitized more than 15 months after origination, corresponding to 10% of volume in the T-Loan dataset. However, such loans would be exceptional because information asymmetries in the post-origination CMBS market led to unravelling in the early 2000s (e.g., An et al., 2011). I thank an anonymous referee for pointing this out.

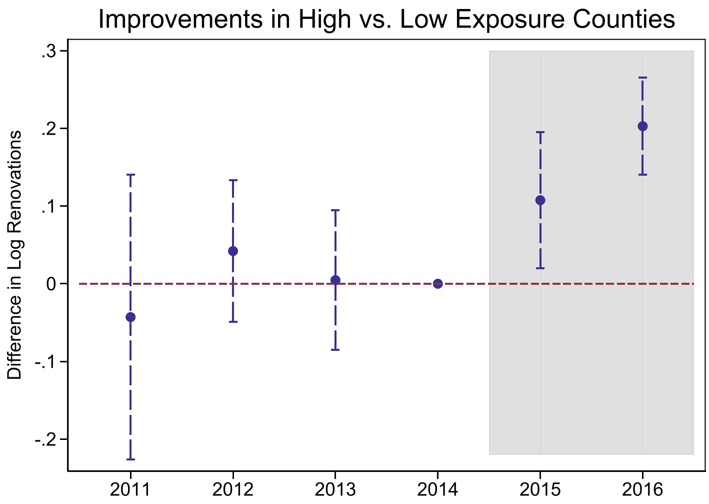


Fig. 4. County-level quality improvements around the HVCRE regulatory shock. *Note:* This figure plots the year-by-year average difference in log renovated properties between counties with high and low values of the exposure variable, which allows for an inspection of pre-trends in county-level quality improvement activity. The reference year is 2014. The exposure variable is banks' share of apartment loan balances in 2010, corresponding to the variable *Total Bank Share*$_c$. High and low are defined according to the average across counties. The difference is demeaned by a state-year fixed effect, as in Table 2. The gray region highlights the period when HVCRE regulations are in place. Brackets are a 95% confidence interval with standard errors clustered by county and year. Data for the variable on the vertical axis are from Trepp's T-Loan dataset. Data for the exposure variable are from Trepp's T-ALLR and T-Loan datasets.

market are not predisposed to non-HVCRE shocks to improvement activity that coincide with the introduction of HVCRE regulation. Measuring *Bank Share*$_c$ in terms of balances helps support the validity of this assumption, since, unlike originations, balances reflect expectations that were formed longer in the past. Moreover, the county fixed effect α_c absorbs slow-moving, county-specific factors that encourage improvement activity (e.g., geography), while the year fixed effect α_t absorbs aggregate factors affecting all counties at the same time (e.g., interest rates).

I devote Section 6 to assessing the validity of assumption (5). However, as a first pass, Fig. 4 provides evidence that the assumption is not violated because of a pre-trend in quality improvement activity. The figure plots the year-by-year average difference in log renovated properties between counties with above and below-average values of *Total Bank Share*$_c$. The two sets of counties exhibit parallel trends leading up to the HVCRE shock, after which quality improvement activity increases significantly in counties with above-average exposure. This finding supports the validity of assumption (5).

5.2. Baseline results

Table 2 reports the results from estimating Eq. (1). I begin by estimating Eq. (1) using the restricted set of counties and outcome variables observed in the T-ALLR dataset, where regulatory incentives are encoded most strongly. The results in column 1 imply that counties with a higher value of *Portfolio Bank Share*$_c$ see a significant post-HVCRE increase in bank loans for non-construction purposes, a proxy for improvement activity given that I do not directly observe improvements in the T-ALLR dataset. This specification is restrictively pure because it relies exclusively on portfolio loans, but it nonetheless implies that the HVCRE shock increases quality improvement activity. In the remainder of the table, I relax these restrictions on sample and outcome variables.

The outcome in columns 2–4 is the log number of renovated properties, based on the T-Loan dataset. The results across these columns imply that improvement activity increases after the introduction of HVCRE regulation in counties with a higher level of exposure, as measured by *Portfolio Bank Share*$_c$ and *Securitized Bank Share*$_c$ separately. Consistent with theory, the results appear to be driven by exposure to banks who originate portfolio loans (i.e., *Portfolio Bank Share*$_c$). The point estimates are similar with or without county controls or state-year fixed effects, but including these additional terms reduces the standard error because they absorb some of the residual variation.

In the remaining columns, I assess the magnitude of the effect by using the composite exposure measure, *Total Bank Share*$_c$. To interpret the point estimate in column 5, counties with a 10 pps higher initial bank share see around a 3.2% increase in renovated properties after the introduction of HVCRE regulation. In column 6, I study log number of renovated apartments. The result is qualitatively similar to its counterpart in column 5, and the larger point estimate likely reflects economies of scale that incentivize improvements on larger properties. Finally, in column 7, I study log total dollar revenue of renovated apartments and obtain a consistent result. Across columns, the point estimates are large enough to imply significant aggregate effects, as I will discuss in Section 8.

In summary, this section shows how an increase in the supply of bank-intermediated financing induced by HVCRE regulation increases real quality improvement activity over 2015–16, thereby exchanging less-expensive housing with more-expensive housing. I assess the internal validity of these results in the next section.

Table 2
The supply of financing and quality improvement activity.

Outcome:	$\log(\textit{Improvement Measure}_{c,t})$						
Measure:	*Bank Loans* (1)	*Renovated Properties* (2)	*Renovated Properties* (3)	*Renovated Properties* (4)	*Renovated Properties* (5)	*Renovated Apartments* (6)	*Renovation Value* (7)
$\textit{Portfolio Bank Share}_c \times \textit{Post}_t$	0.887	0.516	0.518	0.401			
	(0.014)	(0.083)	(0.040)	(0.030)			
$\textit{Securitized Bank Share}_c \times \textit{Post}_t$		0.196	0.245	0.278			
		(0.105)	(0.041)	(0.008)			
$\textit{Total Bank Share}_c \times \textit{Post}_t$					0.318	1.611	3.163
					(0.002)	(0.008)	(0.005)
Sample	Portfolio Loan	Full	Full	Full	Full	Full	Full
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	No	No	Yes	Yes	Yes	Yes	Yes
State-Year FE	No	No	No	Yes	Yes	Yes	Yes
R-squared	0.514	0.571	0.612	0.720	0.720	0.693	0.694
Number of Observations	406	3,128	3,128	3,128	3,128	3,128	3,128

Note: P-values are in parentheses. This table estimates Eq. (1), which is the paper's baseline equation. Subscripts c and t denote county and year. The regression equation is of the form

$$Y_{c,t} = \beta(\textit{Bank Share}_c \times \textit{Post}_t) + \alpha_c + \alpha_t + \gamma X_{c,t} + u_{c,t},$$

where observations are county-years; $\textit{Bank Share}_c$ is a measure of banks' share of apartment loan balances in 2010; and $\textit{Post}_t$ indicates whether t is greater than or equal to 2015. The three measures of $\textit{Bank Share}_c$ are: bank portfolio loan balances divided by the sum of bank portfolio loan, bank securitized loan, and nonbank securitized loan balances, denoted $\textit{Portfolio Bank Share}_c$; bank securitized loan balances divided by the same denominator, denoted $\textit{Securitized Bank Share}_c$; and the sum of bank portfolio and securitized loan balances divided by the same denominator, denoted $\textit{Total Bank Share}_c$. The outcome is the log of a measure of improvement activity: column 1 uses the number of bank portfolio loans originated for purposes other than construction; columns 2–5 use the number of renovated apartment properties; column 6 uses the number of renovated apartment units; and column 7 uses total revenue of renovated apartment units. Column 1 restricts the sample and outcome variable to those observed in Trepp's T-ALLR dataset, which derives exclusively from bank portfolio loans, and the remaining columns are based on the combination of Trepp's T-ALLR and T-Loan datasets. County controls are log real income per capita for the surrounding MSA, log winter storms, and the principal-weighted averages of the following characteristics of existing securitized apartment loans: loan-to-value ratio, debt service coverage ratio, adjustable rate mortgage share, and share of 60+ day delinquent loans. There are 113 and 554 counties in column 1 and columns 2–7, respectively. Observations are weighted by the number of apartments in the county. The sample period is 2011–16. Standard errors are clustered by county. Data for the outcome and exposure variables are from Trepp's T-ALLR and T-Loan datasets. Table 1 lists each variable's data source.

6. Robustness

I perform multiple tests in this section to evaluate: sensitivity to data provider, the exclusion restriction in Eq. (5), and the strength of the shock's first-stage effect on lending behavior. The results of all these tests support the validity of research design described in the previous section. Establishing such validity is important, since I will again use this research design in Section 7 to assess how the HVCRE-induced increase in improvement activity affects housing affordability. Appendix B contains additional robustness tests.

6.1. Sensitivity to data provider

I assess the baseline results' sensitivity to the use of Trepp's data by reperforming my core analysis using the RCA dataset. As mentioned in Section 2, I observe renovations in the RCA dataset and so can calculate $\textit{Renovated Properties}_{c,t}$, with the caveat that the resulting variable will omit renovations on non-transacting properties. Moreover, because the RCA dataset includes loans originated by both bank and nonbank lenders, I can calculate the exposure variable, $\textit{Total Bank Share}_c$. Importantly, the RCA dataset includes both portfolio and securitized loans, and so, as with the core Trepp dataset, the resulting exposure variable will fully encode the regulatory incentives generated by HVCRE regulation.

In columns 1–5 of Table 3, I estimate specifications analogous to those in columns 1–5 of the baseline Table 2 after calculating $\textit{Renovated Properties}_{c,t}$ using the RCA dataset. As in the baseline table, the results imply that counties more exposed to HVCRE regulation see a significant increase in quality improvement activity, and the point estimate in column 5 is almost identical to its analogue from column 5 of the baseline table. Also consistent with the baseline results, the results appear to be driven by exposure to bank portfolio lending. To reiterate, the renovations observed in the RCA dataset occur on properties financed by both portfolio and securitized loans, which means that the baseline results in Table 2 cannot be driven by the fact that $\textit{Renovated Properties}_{c,t}$ is calculated using the T-Loan dataset in that table.

In column 6 of Table 3, I calculate both the outcome and exposure variables using the RCA dataset and find a similar result as in the analogous specification from column 5 of the baseline Table 2. Finally, in column 7 of Table 3, I calculate the outcome variable using the Trepp dataset (i.e., T-Loan) and the exposure variable using the RCA dataset, and again I find a similar result as in the baseline table. That the RCA-calculated exposure variable can also explain Trepp-calculated renovations provides strong

Table 3
Robustness to data provider.

Outcome:	$\log\left(\textit{Renovated Properties}_{c,t}\right)$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\textit{Portfolio Bank Share}_c \times \textit{Post}_t$	1.432	1.302	1.291	1.034			
	(0.001)	(0.009)	(0.000)	(0.000)			
$\textit{Securitized Bank Share}_c \times \textit{Post}_t$		−0.206	−0.109	−0.054			
		(0.369)	(0.588)	(0.759)			
$\textit{Total Bank Share}_c \times \textit{Post}_t$					0.317	0.407	0.204
					(0.097)	(0.001)	(0.008)
Source of Outcome Variable	RCA	RCA	RCA	RCA	RCA	RCA	Trepp
Source of Exposure Variable	Trepp	Trepp	Trepp	Trepp	Trepp	RCA	RCA
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	No	No	Yes	Yes	Yes	Yes	Yes
State-Year FE	No	No	No	Yes	Yes	Yes	Yes
R-squared	0.703	0.704	0.751	0.822	0.816	0.819	0.728
Number of Observations	2,721	2,721	2,721	2,716	2,716	2,117	2,117

Note: P-values are in parentheses. This table estimates Eq. (1) using data from two separate providers to measure the outcome and exposure variables. Subscripts c and t denote county and year. The specifications are similar to those in Table 2. The outcome variable is defined as in Table 2 and measured using the Real Capital Analytics (RCA) dataset in columns 1–6 and using Trepp's T-Loan dataset in column 7. The exposure variables are defined as in Table 2 and measured using Trepp's T-ALLR and T-Loan datasets in columns 1–5 and using the RCA dataset in columns 6–7. The remaining notes are the same as in Table 2.

support for the Trepp-calculated exposure variable used in my core analysis.

Summarizing, the similarity of results across datasets strongly supports the validity of the core research design. In particular, the results of this exercise imply that the baseline findings are not biased due to sample selection in either the T-ALLR or T-Loan components of the core Trepp dataset.

6.2. Exposure to government sponsored enterprises

The Government Sponsored Enterprises (GSEs) purchase almost 50% of apartment loans and, therefore, play a critical role in this market (e.g., Passy (2019)). Moreover, federal regulations affecting the GSEs' behavior changed over the period of analysis, with, for example, a relaxation of apartment lending caps in 2016. The baseline results could be biased upward if counties more exposed to HVCRE regulation are also more exposed to an expansion of GSE-financed improvement loans.[9]

To assess this possibility, I reestimate Eq. (1) after controlling for the share of securitized loan balances backed by the GSEs, denoted $\textit{GSE Share}_{c,t}$. This variable absorbs the effect of changes in the GSEs' behavior. If indeed such changes drive the baseline estimates, then the estimated coefficient on $\textit{Total Bank Share}_c \times \textit{Post}_t$ should be close to zero. However, the resulting coefficient of interest in Table 4 is similar to its analogue from the baseline Table 2. This finding suggests that the baseline results are not biased due to changes in the GSEs' behavior, thereby supporting their validity.

Table 4
Robustness to GSE exposure.

Outcome:	$\log\left(\textit{Renovated Properties}_{c,t}\right)$		
	(1)	(2)	(3)
$\textit{Total Bank Share}_c \times \textit{Post}_t$	0.318	0.351	0.318
	(0.071)	(0.030)	(0.002)
$\textit{GSE Share}_{c,t}$	0.052	−0.002	−0.004
	(0.260)	(0.964)	(0.944)
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
County Controls	No	Yes	Yes
State-Year FE	No	No	Yes
R-squared	0.567	0.609	0.720
Number of Observations	3125	3125	3125

Note: P-values are in parentheses. This table estimates a variant of Eq. (1) that assesses robustness to changes in lending practices by the Government Sponsored Enterprises (GSEs). Subscripts c and t denote county and year. The specifications are similar to those in Table 2 after controlling for the share of securitized loan balances backed by the GSEs, denoted $\textit{GSE Share}_{c,t}$, based on Trepp's T-Loan dataset. Data for the outcome variable are from Trepp's T-Loan dataset. Data for the exposure variable are from Trepp's T-ALLR and T-Loan datasets. The remaining notes are the same as in Table 2.

6.3. Large borrowers

Securitized apartment loans are typically larger than portfolio loans (e.g., Ghent and Valkanov, 2016), and so the T-Loan dataset may overrepresent large borrowers who are unlikely to face credit supply constraints. Therefore, since I measure many of my outcome variables using the T-Loan dataset, the baseline results may spuriously capture channels orthogonal to the HVCRE shock if they are driven by such borrowers. To evaluate whether such unconstrained borrowers drive the results, I recalculate $\textit{Renovated Properties}_{c,t}$ after excluding renovations by borrowers who obtain credit from more than one distinct source in 2010. Then, I reestimate Eq. (1) and report the results in columns 1–3 of Appendix Table A2. The estimated coefficient of interest is similar to that from the baseline Table 2, which implies that the baseline results are not driven by large, unconstrained borrowers.

[9] I thank an anonymous referee for raising this possibility.

6.4. Loan reclassification

Banks have an incentive to underreport HVCRE lending, per the 2017 Conference of State Bank Supervisors' report. To the extent that banks reclassify "true" construction as an improvement project, the baseline results may overestimate the shock's real effect on quality improvement activity. Since I cannot verify whether a property's foundation has truly changed, thus officially qualifying the project as construction, I assess the scope for such bias by excluding potentially reclassified renovations. I define such renovations as occurring on either very new (i.e., under 3 years old) or very old properties (i.e., built before 1940), in which the property's foundation is more likely to have changed. Reperforming the baseline analysis after excluding such renovations yields similar estimates, as shown in columns 4–6 of Appendix Table A2. Thus, while reclassification may indeed occur, it does not appear to bias the results.

6.5. Constrained demand as an amplification mechanism

Building on the exercise from Section 6.3, quantitative models (e.g., Bernanke and Gertler, 1989) typically require borrowing constraints to generate large real effects. I now assess the role of constrained demand for improvement financing as an amplification mechanism that drives the baseline results. I test this hypothesis by reestimating Eq. (1) after interacting the treatment variable with county characteristics that proxy for borrowing constraints or the level of unconstrained demand for improvement financing.

Appendix Table A3 reports the results of this exercise. Column 1 shows how the effect of the HVCRE shock is stronger where the average borrower obtains credit from fewer distinct sources, a proxy for constraints on her ability to access credit due to information asymmetries. In column 2, the effect is stronger in higher-income counties, which may reflect how higher-income renters are more willing to pay for higher-quality housing (e.g., Jaravel, 2019; Handbury, 2019). Thus, real estate investors (i.e., borrowers) in such counties have higher unconstrained demand for improvement financing, so that an increase in financial supply has larger real effects. Similarly, column 3 implies a stronger effect in the absence of rent stabilization policies. Such policies discourage real estate investors from making improvements by limiting their ability to raise rent, thus reducing investors' unconstrained demand for improvement financing.

Together, the results in Appendix Table A3 suggest that HVCRE regulation relaxes borrowing constraints on real estate investors, per column 1, and it has stronger real effects where these investors' unconstrained demand for improvement financing is higher, per columns 2 and 3. These findings are consistent with the predictions of quantitative models, thereby supporting the validity of the baseline results.

6.6. Confounding regulatory shocks

In addition to capital requirements, the post-2010 period saw the introduction of the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR), both associated with Basel III Accords. However, the absence of a lender-level pre-trend in Fig. 3 makes it unlikely that the results confound these other two regulations. In particular, unlike capital risk weights, the liquidity risk weights associated with the LCR do not vary by project type, and Gete and Reher (2021) show that much of the adjustment to the LCR occurred in 2014. Moreover, the U.S. version of the NSFR was not proposed until May 2016.

Table 5
Robustness to property-level analysis.

Outcome:	*Probability of Renovation*$_{i,\ell,t}$	
	(1)	(2)
$Bank_\ell \times Post_t$	0.012	0.012
	(0.009)	(0.010)
Property-Lender FE	Yes	Yes
County-Year FE	Yes	Yes
Zip Code Controls	No	Yes
R-squared	0.308	0.308
Number of Observations	30,733	30,733

Note: P-values are in parentheses. This table estimates Eq. (6), which is a property-level analogue to the baseline equation. Subscripts i, c, z, ℓ, and t denote property, county, zip code, lender, and year. The regression equation is of the form

$$Probability\ of\ Renovation_{i,\ell,t} = \beta(Bank_\ell \times Post_t) + \alpha_{c(i),t} + \alpha_{i,\ell} + \gamma X_{z(i),t} + u_{i,\ell,t},$$

where observations are property-years; $Bank_\ell$ indicates whether the existing loan on the property is originated by a bank; and $Probability\ of\ Renovation_{i,\ell,t}$ indicates whether a renovation occurs. Zip code controls are log average income, log number of tax returns, and log average rent. There are 6100 properties. The sample period is 2011–16. Standard errors are clustered by property. Data are from Trepp's T-Loan dataset.

6.7. Property-level analysis

Taking advantage of my detailed microdata, I construct a property-level difference-in-difference research design that is analogous to the core, county-level analysis. This approach allows me to include county-year fixed effects so that I can rely on a very weak identification assumption. The regression equation is

$$Probability\ of\ Renovation_{i,\ell,t} = \beta(Bank_\ell \times Post_t) + \alpha_{c(i),t} + \alpha_{i,\ell} + u_{i,\ell,t}, \quad (6)$$

where i, ℓ, and t index properties, lenders, and years; $Bank_\ell$ indicates if the property owner's lender is a bank; $Probability\ of\ Renovation_{i,\ell,t}$ indicates whether a renovation occurs in t; and $c(i)$ denotes the county to which i belongs. The county-year fixed effect $\alpha_{c(i),t}$ absorbs contemporaneous demand shocks, and the property-lender fixed effect $\alpha_{i,\ell}$ limits variation to the same relationship. I use the T-Loan dataset for this exercise because it allows me to calculate all of the variables in Eq. (6). The associated summary statistics are shown in Appendix Table A4.

Table 5 reports the results. The point estimate implies that properties whose owner relies on bank financing have a 1.2 pps higher annual probability of renovation after the introduction of HVCRE regulation. Quantitatively, the effect is equal to 46% of the unconditional property-level probability of 2.6%. These property-level results support the validity of the baseline findings from Section 5.

6.8. Verifying the first-Stage effect

I now verify that HVCRE regulation increases the supply of bank credit for improvement projects, which serves as the "first stage" for my main, county-level analysis in Section 5. All of the exercises in this subsection rely on the T-Loan dataset because it covers properties financed by both bank and nonbank lenders, whose identity I observe. This enables me to construct a lender-level dataset, summarized in Appendix Table A4.

6.8.1. First-stage effect: triple difference-in-difference

First, I estimate a triple difference-in-difference equation that intuitively asks whether lenders more exposed to HVCRE regulation, namely banks, shift their lending from construction to improvement projects more than nonbanks after the policy's introduction. Separating loans by the type of project they finance allows me to include lender-year fixed effects, which absorb confounding shocks to the overall level of lending due to, say, other Dodd-Frank regulations.

I estimate the following equation over 2011–16,

$$Y_{k,\ell,t} = \beta(Bank_\ell \times Post_t \times Renovation_k) + \dots \qquad (7)$$
$$\dots + \gamma(Bank_\ell \times Renovation_k) + \alpha_{\ell,t} + \alpha_{k,t} + u_{k,\ell,t},$$

where k, ℓ, and t index loan purpose, lender, and year; $Bank_\ell$ indicates if the lender is a bank; $Renovation_k$ indicates if the purpose is a renovation, where the set of loan purposes are renovation or construction as described in Appendix A.2; $Y_{k,\ell,t}$ is the log number of loans originated or dollar volume for purpose k; and $\alpha_{\ell,t}$ and $\alpha_{k,t}$ are lender-year and purpose-year fixed effects. The parameter of interest in Eq. (7) is β, which captures the triple difference between treated loan types ($Renovation_k$) originated by treated lenders ($Bank_\ell$) during the treatment period ($Post_t$), and the counterfactual purpose-lender-years.

The results from estimating Eq. (7) are shown in columns 1–2 of Table 6. The point estimate in column 1 implies that banks increase the ratio of improvement to construction loans by 33 log points relative to nonbanks after HVCRE regulation is introduced. The magnitude is larger when studying dollar volume in column 2, which may reflect economies of scale that incentivize improvements on larger properties.

6.8.2. First-stage effect: difference-in-difference

The lender-year effects $\alpha_{\ell,t}$ in Eq. (7) enable me to obtain tightly identified estimates of how HVCRE regulation affects banks' portfolio composition. However, this restrictiveness prohibits inference about whether banks actually finance more improvement projects. Therefore, I next estimate the difference-in-difference equation

$$Y_{\ell,t} = \beta(Bank_\ell \times Post_t) + \alpha_t + \alpha_\ell + \gamma X_{\ell,t} + u_{\ell,t}, \qquad (8)$$

where the notation is similar to that in Eq. (7), although observational units are now lender-years, as opposed to purpose-lender-years.

Columns 3–4 of Table 6 report the results from estimating Eq. (8). My outcome of interest, $Y_{\ell,t}$, is the log number of renovated apartments financed by new loans. The point estimate in column 3 implies that banks finance significantly more improvements relative to nonbanks in the post-HVCRE period, and the results are robust to including lender controls in column 4. Appendix Figure A2 tests for differences in average portfolio characteristics between banks and nonbanks, and the resulting similarity suggests that the estimates in Table 6 are unlikely to be biased because of differences in loan specialization.

Collectively, these findings show how banks increase their improvement lending rather than simply reducing their construction lending, which is consistent with the goal of maintaining the same total exposure to commercial real estate projects.

6.8.3. Robustness of the first-stage effect

In terms of robustness, I first assess whether the first-stage results are biased from undersampling small regional banks who typically originate a large share of construction loans. Such undersampling is possible because construction loans typically have a construction-to-permanent financing structure - where the lender provides a short-term note that converts to a long-term note once the project has stabilized - and these loans are more difficult to securitize prior to conversion (Black et al., 2017). Therefore, I interact the treatment variable in Eq. (8) with the lender's ratio of construction loans to total assets in 2010, normalized to have zero mean and unit variance. Column 1 of Appendix Table A5 shows how the estimated coefficient on this interaction term is positive, suggesting that banks with a focus on construction lending are indeed represented in the T-Loan dataset and that, as expected, their behavior drives the results. In column 2, I drop the Big-4 banks from the sample and obtain almost the same point estimate as when using the full sample in Table 6, further supporting the validity of the first-stage effect.

Next, poorly capitalized banks should theoretically increase their supply of improvement loans by more than well-capitalized banks because HVCRE regulation imposes a more binding constraint on them. I test this theory by reestimating Eq. (8) after interacting the treatment variable with the ratio of total equity to total assets in 2010, normalized to have zero mean and unit variance. Consistent with theory, I obtain a negative estimated coefficient on the interaction term, as shown in column 3 of Appendix Table A5. This consistency with theory supports the validity of the first-stage effect.

Finally, columns 4–5 of Appendix Table A5 show that the price of credit for bank-originated improvement loans also falls after the introduction of HVCRE regulation, consistent with a movement along the credit demand curve. The modest price response may reflect a substitution toward higher-risk improvement projects or binding borrowing constraints. Using a different methodology, Glancy and Kurtzman (2018) also find that HVCRE regulation modestly affects interest rates.

Table 6
Robustness of first-stage effect on lending behavior.

Specification:	Triple Difference-in-Difference		Difference-in-Difference	
Outcome:	$\log\left(Loans_{k,\ell,t}\right)$ (1)	$\log\left(Volume_{k,\ell,t}\right)$ (2)	$\log(Renovations_{\ell,t})$ (3)	(4)
$Bank_\ell \times Post_t \times Renovation_k$	0.332 (0.000)	5.385 (0.017)		
$Bank_\ell \times Post_t$			1.210 (0.024)	1.141 (0.030)
Lender-Year FE	Yes	Yes		
Purpose-Year FE	Yes	Yes		
Bank × Renovation	Yes	Yes		
Lender FE			Yes	Yes
Year FE			Yes	Yes
Lender Controls			No	Yes
R-squared	0.762	0.799	0.660	0.667
Number of Observations	966	966	582	582

Note: P-values are in parentheses. This table estimates Eqs. (7) and (8), which assess the lender-level effect of HVCRE regulation as a first stage. Subscripts k, ℓ and t denote loan purpose, lender, and year. Columns 1–2 estimate Eq. (7),

$$Y_{k,\ell,t} = \beta(Bank_\ell \times Post_t \times Renovation_k) + \gamma(Bank_\ell \times Renovation_k) + \alpha_{\ell,t} + \alpha_{k,t} + u_{k,\ell,t},$$

where observations are purpose-lender-years; $Bank_\ell$ denotes if lender ℓ is a bank; $Renovation_k$ indicates whether the purpose is a renovation; and the set of loan purposes are renovation or construction. $Loans_{k,\ell,t}$ is the number of loans for purpose k originated by ℓ in t, and $Volume_{k,\ell,t}$ is the corresponding dollar volume. Columns 3–4 estimate Eq. (8),

$$Y_{\ell,t} = \beta(Bank_\ell \times Post_t) + \alpha_t + \alpha_\ell + \gamma X_{\ell,t} + u_{\ell,t},$$

where observations are lender-years. $Renovations_{\ell,t}$ is the number of renovated apartments financed by a new loan by lender ℓ in t. Lender controls are principal-weighted averages of the following characteristics of existing loans: loan-to-value ratio, debt service coverage ratio, adjustable rate mortgage share, and share of delinquent loans. There are 122 lenders. Observations are weighted by the lender's market share. The sample period is 2011–16. Standard errors are clustered by lender and year in columns 1–2 and by lender in columns 3–4. Data are from Trepp's T-Loan dataset.

7. Implications for housing affordability

Since quality improvements reduce the supply of relatively inexpensive housing units by transforming them into relatively expensive ones, the baseline results have direct implications for policy discussion about housing affordability. I now make these implications explicit by estimating how the HVCRE shock affects the supply of apartments and rent across the quality distribution. To do so, I use my baseline difference-in-difference setup, the validity of which is strongly supported by the robustness tests in Section 6 and Appendix B.

7.1. Supply of apartments

The HVCRE shock affects the supply of apartments through two channels. First, by increasing the supply of financing for improvement projects, which, by definition, transform low-quality apartments into high-quality ones, the shock affects the distribution of apartment supply across quality segments (i.e., "the improvement channel"). This channel is my primary focus, given the surge in improvement activity documented in Section 3. However, HVCRE regulation increases the supply of improvement financing by reallocating resources away from construction projects. Therefore, the shock also affects the overall level of apartment supply (i.e., "the construction channel"). I test these hypotheses by reestimating my baseline regression Eq. (1) after replacing the outcome with a measure of the change in apartment supply.

Table 7 reports the results. In columns 1–3, I verify that the HVCRE shock reduces county-level log apartment construction and, thus, growth in overall apartment supply. This finding confirms the construction channel described in the previous paragraph. In columns 4–6, I replace the outcome with the change in the log number of apartment properties ranked in the top quality segment by professional property inspectors. The results imply that the shock increases growth in the supply of high-quality apartments, even as it reduces growth in overall supply. Both effects appear to be driven by exposure to bank portfolio loans, as suggested by columns 3 and 6.

Taken together, these two sets of results are surprising because, in practice, most new construction occurs in the upper segments of the quality distribution (Rosenthal, 2014). Thus, the improvement channel more than offsets the construction channel, leading to a positive effect on the supply of rental housing quality. This positive net effect likely stems from the fact that it costs less to improve a property than to build a new one (RSMeans Company LLC, 2014).

Table 7
Implications of quality improvement shock for the supply of apartments.

Outcome:	$\log\left(\textit{Newly-Built Properties}_{c,t}\right)$			$\Delta\log\left(\textit{Top-Quality Properties}_{c,t}\right)$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\textit{Total Bank Share}_c \times \textit{Post}_t$	−0.427	−0.404		0.578	0.612	
	(0.017)	(0.023)		(0.005)	(0.003)	
$\textit{Portfolio Bank Share}_c \times \textit{Post}_t$			−0.532			0.811
			(0.098)			(0.004)
$\textit{Securitized Bank Share}_c \times \textit{Post}_t$			−0.323			0.486
			(0.107)			(0.018)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	No	Yes	Yes	No	Yes	Yes
R-squared	0.556	0.559	0.559	0.117	0.125	0.126
Number of Observations	3,128	3,128	3,128	2,530	2,530	2,530

Note: P-values are in parentheses. This table estimates a variant of Eq. (1) that assesses how the HVCRE shock affects the supply of apartments. Subscripts c and t denote county and year. The specifications are similar to those in Table 2 with a different outcome variable that measures growth in the supply of apartment properties: $\log\left(\textit{Newly-Built Properties}_{c,t}\right)$ is the log of number of apartment construction projects; and $\Delta\log\left(\textit{Top-Quality Properties}_{c,t}\right)$ is the change in the log of number of properties times share of apartments ranked in the top quality segment by professional property inspectors. Data for the outcome variables are from Trepp's T-ALLR and T-Loan datasets and other datasets shown in Table 1. Data for the exposure variables are from Trepp's T-ALLR and T-Loan datasets. The remaining notes are the same as in Table 2.

Table 8
Implications of quality improvement shock for housing affordability.

Outcome:	$\Delta\log\left(\textit{Average Rent}_{c,t}\right)$			$\Delta\textit{Cost-Burdened Share}_{c,t}$			$\Delta\log\left(\textit{Top-Quintile Rent}_{c,t}\right)$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\textit{Total Bank Share}_c \times \textit{Post}_t$	0.043	0.044		0.072	0.072		−0.048	−0.051	
	(0.024)	(0.013)		(0.003)	(0.003)		(0.096)	(0.076)	
$\textit{Portfolio Bank Share}_c \times \textit{Post}_t$			0.036			0.078			−0.070
			(0.120)			(0.011)			(0.013)
$\textit{Securitized Bank Share}_c \times \textit{Post}_t$			0.050			0.068			−0.040
			(0.006)			(0.007)			(0.185)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
R-squared	0.241	0.262	0.262	0.366	0.374	0.374	0.280	0.290	0.292
Number of Observations	2,480	2,480	2,480	2,565	2,565	2,565	2,003	2,003	2,003

Note: P-values are in parentheses. This table estimates a variant of Eq. (1) that assesses how the HVCRE shock affects various outcomes related to housing affordability. Subscripts c and t denote county and year. The specifications are similar to those in Table 2 with a different outcome variable: $\Delta\log\left(\textit{Average Rent}_{c,t}\right)$ is the change in the log average rent among all apartments; $\Delta\textit{Cost-Burdened Share}_{c,t}$ is the change in the share of apartments with rent greater than 30% of the zip code's average income; and $\Delta\log\left(\textit{Top-Quintile Rent}_{c,t}\right)$ is the change in the log average rent among apartments in the top 20% by level of rent relative to the county-year. A rent-to-income ratio above 30% is the standard threshold for being considered cost-burdened (JCHS, 2017). Columns 7–9 also include a vector of state-year fixed effects. Data for the outcome variables are from Trepp's T-Loan dataset and other datasets shown in Table 1. Data for the exposure variables are from Trepp's T-ALLR and T-Loan datasets. The remaining notes are the same as in Table 2.

7.2. Housing affordability

Turning more directly to the question of housing affordability, I reestimate Eq. (1) after replacing the outcome variable with various measures of rental housing costs. The results in columns 1–3 of Table 8 show how counties more exposed to the shock see significantly higher rent growth on the average apartment. This finding reflects a mixture of higher average quality (i.e., the improvement channel) as well as lower overall supply (i.e., the construction channel). It is beyond this paper's scope to separate these two channels, but the results in columns 7–9, discussed shortly, will support the empirical relevance of the improvement channel.

In columns 4–6, I replace the outcome with the change in the share of cost-burdened renters, conventionally defined as having a rent-to-income ratio above 30% (JCHS, 2017). The results show how counties more exposed to the HVCRE shock see significant growth in the share of cost-burdened renters. Importantly, I use the term "cost-burdened" to match the language typically heard in policy discussions about housing affordability (e.g., Donovan, 2014), but the true welfare content of this section's results is in fact quite complicated, as I discuss in my conclusion.

Lastly, the outcome in columns 7–9 is rent growth among high-quality apartments, here measured as the most expensive quintile of apartments in the county-year. The negative estimated coefficients imply that the HVCRE shock reduces rent growth among such apartments, which, along with the results in columns 4–6 of Table 7, is consistent with a movement along the demand curve for housing quality. While the point estimates are only significant at the 10% level when using the composite exposure variable in columns 7–8, there is a highly significant effect when focusing on exposure to bank portfolio loans, shown in column 9. As in previous tables, this finding is consistent with the stronger regulatory incentives associated with such loans.[10]

Summarizing, HVCRE regulation increases the average apartment's rent growth, but it reduces rent growth on high-quality apartments. These findings are consistent with the research hypothesis outlined in Section 3.1: an increase in the supply of improvement financing raises average housing quality and, thus, average rent, but high-quality rent must fall in order for households to accommodate this increase in the supply of housing quality.

8. Aggregate implications

I conclude by assessing the aggregate effect of credit supply on quality improvement activity and, subsequently, on rent growth. Consider a counterfactual without HVCRE regulation, in which capital requirements treat loans for all residential investment projects equally. I first ask how many fewer apartments would have been renovated under this counterfactual, expressed as a share of actual renovations and denoted η. In other words, η is the share of observed renovations attributable to HVCRE regulation. I introduce two additional assumptions to calculate this statistic.

Assumption 1 (Control Group). The effect of the HVCRE shock on quality improvement activity is zero in counties whose value of $\textit{Total Bank Share}_c$ is below $P_B(\textit{Total Bank Share}_c)$, where $P_B(\textit{Total Bank Share}_c)$ denotes the Bth percentile of $\textit{Total Bank Share}_c$ across counties. These counties are defined as the control group.

I introduce Assumption 1 because I do not know how small the exposure variable, $\textit{Total Bank Share}_c$, must be in order for a county to be effectively unexposed to HVCRE regulation (i.e., in the "control group"). Assumption 1 defines this threshold as the Bth percentile of $\textit{Total Bank Share}_c$ across counties. Consequently, I can write the effect of HVCRE regulation on county c as

$$\eta_c = 1 - \exp\left[-\beta \times \max\left\{\textit{Total Bank Share}_c - P_B(\textit{Total Bank Share}_c), 0\right\}\right]. \tag{9}$$

The least conservative approach would be to define the control group as counties with $\textit{Total Bank Share}_c = 0$, corresponding to $B = 0\%$ in my data (i.e., the zeroth percentile). Therefore, following the literature (e.g., Chodorow-Reich, 2014), I report results for B between 5% and 10%.

The second assumption relates to partial equilibrium. HVCRE regulation may affect quality improvement activity through general equilibrium forces, but these effects are subsumed by the year fixed effect in Eq. (1). It is beyond this paper's scope to quantify general equilibrium effects, and so I introduce the following assumption, which effectively states that general equilibrium effects are sufficiently small that the aggregate effect equals the sum of county-level effects.

Assumption 2 (Partial Equilibrium). The effect of the HVCRE shock on aggregate quality improvement activity is equal to the weighted sum of county-level effects, η_c. In particular, the aggregate share of renovated apartments due to the HVCRE shock is

$$\eta = \frac{\sum_c \sum_{t\geq 2015} \eta_c \times \textit{Renovated Apartments}_{c,t}}{\sum_c \sum_{t\geq 2015} \textit{Renovated Apartments}_{c,t}}. \tag{10}$$

Table 9 summarizes the results of this aggregation exercise. Focusing on column 1, HVCRE regulation accounts for between 19% and 24% of quality improvement activity over 2015–16, depending on the definition of the control group. To put the magnitudes in perspective, the property-level analysis from Section 6.7 finds that HVCRE regulation increases a property's annual probability of renovation by 46%. This comparison suggests that the results in column 1 are plausible and do not stem from aggregation assumptions, which are not necessary in the property-level analysis.

Finally, I reweight the raw statistic in Eq. (10) by a factor of 0.7 to reflect the fact that only 70% of apartment renovations occur on properties with a lien, according to the Rental Housing Finance Survey (RHFS). Reweighting the raw statistic helps address the fact that I can only calculate an in-sample aggregate effect.

In column 2, I summarize an analogous statistic in terms of growth in average rent. The results imply that HVCRE regulation accounts for between 25% and 32% of rent growth over 2015–16. As discussed in Section 7.1, this effect reflects a mixture of higher average quality as well as lower overall supply.

8.1. Robustness of the aggregate effect

I assess the plausibility of the large estimates in Table 9 by using the AHS dataset to perform a hedonic quality adjustment. My goal is to assess the extent to which quality improvements have been priced into rent growth. To do so, I construct a hedonic rent index that remeasures rent after correcting for the contribution of both large-scale quality improvements (e.g., adding a floor) and small-scale ones (e.g., adding a dishwasher). The resulting difference between observed and quality-adjusted rent growth reflects the value of quality improvements. I use the AHS dataset for this exercise because it contains a much richer set of property characteristics than the core Trepp datasets. The drawback to the AHS dataset is that I can only perform this exercise over the 2007–13 period because of the AHS's redesign in 2015, mentioned in Section 2. Therefore, this exercise does not directly inform whether the HVCRE-induced improvements contribute to rent growth, but, rather, whether improvements of any origin explain a sufficiently large share of rent growth following the Great Recession to make the estimated effects of HVCRE regulation in Table 9 plausible.

The logic of a hedonic adjustment is to hold the cross-sectional distribution of housing quality fixed and ask how

[10] This finding also explains why the estimated coefficient on $\textit{Portfolio Bank Share}_c$ in column 3 has the correct sign and an economically significant magnitude despite a p-value of 0.12: exposure to bank portfolio loans attenuates average rent growth by lowering rent growth on high-quality apartments.

Table 9
Share of quality improvements and rent growth due to the shock.

Outcome:	Renovated Apartments	Rent Growth
Share of Outcome due to Shock, $B = 5\%$	24%	32%
Share of Outcome due to Shock, $B = 10\%$	19%	25%
Source of β:	Table 2, Column 6	Table 8, Column 2

Note: This table shows the implied contribution of HVCRE regulation to the aggregate value of the indicated outcome variable over 2015–16, denoted η in the text and expressed as a share of the actual value of the outcome variable over 2015–16. The implied effect is based on Assumption 1 (Control Group) and Assumption 2 (Partial Equilibrium). The parameter B indexes the set of counties defined to be in the control group and, thus, not affected by HVCRE regulation. The first and second rows respectively define the control group as counties where banks' initial share of apartment loan balances (i.e., $\textit{Total Bank Share}_c$) is below the fifth or tenth percentile across counties, respectively. Explicitly, letting $P_B(\textit{Total Bank Share}_c)$ denote the Bth percentile of $\textit{Total Bank Share}_c$ across counties, the expression for the statistic shown in column 1 is

$$\eta_1 = \frac{\sum_c \sum_{t\geq 2015} \left(1 - \exp\left[-\beta \times \max\left\{\textit{Total Bank Share}_c - P_B(\textit{Total Bank Share}_c), 0\right\}\right]\right) \times \textit{Renovated Apartments}_{c,t}}{\sum_c \sum_{t\geq 2015} \textit{Renovated Apartments}_{c,t}},$$

and the expression for the statistic in column 2 is

$$\eta_2 = \frac{\sum_c \sum_{t\geq 2015} w_c \times \beta \times \max\left\{\textit{Total Bank Share}_c - P_B(\textit{Total Bank Share}_c), 0\right\}}{\sum_c \sum_{t\geq 2015} w_c \times \Delta \log\left(\textit{Average Rent}_{c,t}\right)},$$

where c and t index counties and years; and w_c is the number of apartments in county c. I multiply the previous two expressions by a factor of 0.70 to reflect the fact that only 70% of improvements occur on apartments with a lien, according to the 2015 RHFS. Data for the outcome variables are from Trepp's T-Loan dataset.

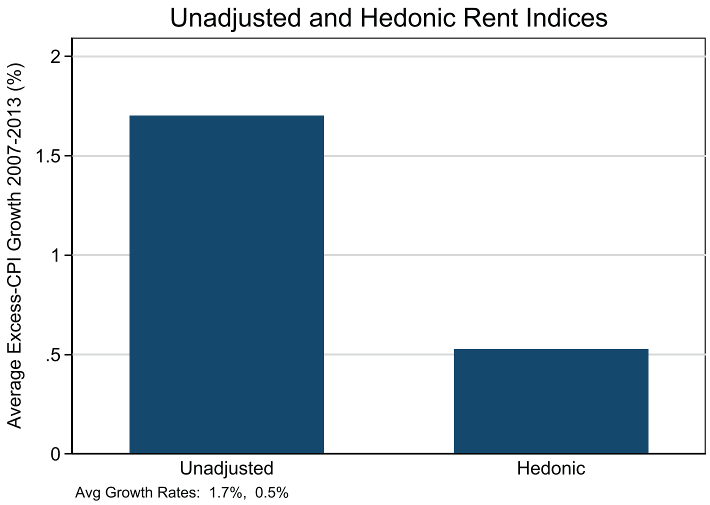


Fig. 5. Magnitude of the quality improvement effect through a hedonic adjustment. *Note:* This figure plots the results of the hedonic quality adjustment from Section 8.1, which assesses the plausibility of the implied aggregate effect of HVCRE regulation from Section 8. The vertical axis shows average annual growth in real (i.e., excess-CPI) rent over 2007–13 for various rent indices. Unadjusted denotes average observed rent. Hedonic denotes the hedonic index defined in Section 8.1. Data are from the Census' AHS dataset.

the average rent in this distribution has grown over time. Thus, the notion of quality-adjusted rent is the expenditure required to live in a housing unit with the same set of structural features. Since my interest is in quality improvements to a given housing unit, I estimate the following pricing kernel in differences

$$\Delta \log\left(\textit{Rent}_{i,t}\right) = \beta^{\Theta} \Delta\Theta_{i,t} + \alpha_i + \alpha_t + u_{i,t}, \tag{11}$$

where i and t index housing units and years; $\Delta \log\left(\textit{Rent}_{i,t}\right)$ is the change in log rent; and $\Delta\Theta_{i,t}$ is a vector of indicators for the installment of features $\theta_{i,t} \in \Theta_{i,t}$, summarized in Appendix Table A6. Note that, like most hedonic pricing kernels, Eq. (11) does not take a stance on the shocks generating variation in improvements, $\Delta\Theta_{i,t}$.

Given the estimates of Eq. (11), shown in Appendix Table A7, I compute a unit's quality-adjusted rent as

$$\textit{Rent}^H_{i,t} = \textit{Rent}_{i,t_0} \times \exp\left[\sum_{\tau=t_0+2}^{t} \left(\Delta \log\left(\textit{Rent}_{i,\tau}\right) - \beta^{\Theta}\Delta\Theta_{i,\tau}\right)\right], \tag{12}$$

where t_0 is the initial year; and the summation begins at t_0+2 because the AHS is biennial. Finally, I aggregate $\textit{Rent}^H_{i,t}$ across housing units as described in Appendix C to calculate the hedonic index.

Fig. 5 summarizes annual growth in unadjusted rent and the hedonic index over the 2007–13 period. Beginning on the left, unadjusted rent grows 1.7% in real terms (i.e., excess of non-housing inflation). Moving to the right, quality-adjusted real rent growth is 0.5%, based on the hedonic index. By extension, quality improvements account for 70% (i.e., $0.70 = \frac{1.7-0.5}{1.7}$) of observed real rent growth. This finding is robust to alternative specifications of Eq. (11) described in Appendix C, and it appears to be unique to the period after the Great Recession, as shown in Appendix Figure A5.

The results of this hedonic adjustment imply that quality improvements - regardless of the shock generating them - explain a significant share of observed real rent growth following the Great Recession. This finding implies makes it plausible that the HVCRE shock and the improvements generated by it explain a large share of observed rent growth over my period of analysis, per the estimates in Table 9.

9. Conclusion

This paper shows how financial intermediaries have contributed to a newly documented surge in improvements to rental housing quality since the Great Recession, and, in so doing, have meaningfully affected housing affordability. Using the introduction of Dodd-Frank bank capital requirements as an exogenous source of variation, I find that a reallocation of bank credit toward improvements accounts for 24% of improvement activity over 2015–16. By increasing the supply of rental housing quality, this shock raises average rent growth but reduces rent growth on high-quality housing. These findings support the conclusion that financial intermediaries supply housing quality by providing financing for improvement projects, and, through their role as suppliers of quality, so affect housing affordability.

From the policy perspective, these findings exemplify an unintended, economically significant consequence of regulations that affect financial intermediaries' portfolio choice. In particular, the results show how housing quality and affordability are affected not only by traditional urban policies (e.g., rent stabilization), but also by regulation of the financial sector. Moreover, from the standpoint of urban policymakers seeking to expand the supply of affordable housing, the results imply that limiting market-rate construction actually leads to upward filtering that makes affordable housing more scarce.

However, the welfare content of this regulatory spillover is unclear. On the one hand, the shock appears to relax borrowing constraints on improvement financing, implying a Pareto improvement. On the other hand, by channeling resources toward improvements and away from construction, the shock may reduce allocative efficiency and be Pareto-impairing. In distributional terms, the increase in the supply of housing quality likely favors higher-income renters, although I leave a formal welfare analysis for future work.

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